Unit distances among vertices of a convex polygon: a bounded-degree reformulation, neighbourhood structure, and the limits of quasi-planarity

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Abstract

Let P be a finite set of n points in strictly convex position in $\mathbb{R}^2$, let $G(P)$ be the graph joining pairs of points at distance exactly 1, and let $U_c(n)$ be the maximum number of such pairs, over all convex n-point sets. Erdős and Moser asked in 1959 whether $U_c(n) = O(n)$; the problem is open, and the best known bounds are $2n-7 \le U_c(n) \le n\log_2 n + 4n$ (Edelsbrunner–Hajnal 1991; Aggarwal 2015). We do not solve the problem. Instead we prove a group of structural results about the unit-distance graphs of convex point sets. The two main ones are: *(i)* the Erdős–Moser conjecture is *equivalent* to the assertion that every finite convex point set contains a vertex incident to at most c unit segments for a single absolute constant c (so the class of graphs $G(P)$ has bounded degeneracy), which reduces the 1959 question to a purely local statement; and *(ii)* for every k and every $n \ge 2k$ there is a convex n-gon whose unit-distance graph contains k pairwise crossing edges, which rules out obtaining the linear bound from any quasi-planarity theorem of bounded order. We also show that $G[N(v)]$ is a disjoint union of paths of at most three vertices for every vertex v, and that unit-degree is bounded by $2kd$ when the points lie on k algebraic curves of degree $\le d$ (in particular $\le 2n$ unit distances for points on a parabola or an ellipse). All constructions are certified by exact rational arithmetic; the code and its tests are included with the paper.

1 Introduction and statement of the problem

In 1946 Erdős asked for the maximum number of times the same distance can occur among n points in the plane; in 1959 Erdős and Moser [1] posed the convex variant that we study. Throughout, a point set is in *strictly convex position* if its points are the vertices of a strictly convex polygon; equivalently every point is an extreme point of the convex hull and no three are collinear.

Definition 1. For a finite $P \subset \mathbb{R}^2$ in strictly convex position write

$$u(P) := \#\big\{\{x,y\} \subseteq P : x \ne y,\ |x-y| = 1\big\}, \qquad U_c(n) := \max\{u(P) : |P| = n,\ P \text{ strictly convex}\},$$

and let $G(P)$ be the *unit-distance graph* on vertex set P with $xy \in E(G(P))$ iff $|x - y| = 1$.

Conjecture 1 (Erdős–Moser 1959, [1]; Erdős problem 96)**.** $U_c(n) = O(n)$.

Erdős and Moser originally guessed $U_c(n) \sim \frac{5}{3}n + O(1)$; this was disproved by Edelsbrunner and Hajnal [2], who constructed convex n-gons with $2n - 7$ unit distances. Erdős and Fishburn

then conjectured the sharper bound $u(P) \le 2n$ [5, 9], which is still open. On the upper-bound side, Füredi [3] proved $U_c(n) \le 12n \log_2 n - 6n$ using 0-1 "cut matrices", Brass and Pach [4] gave a short proof of the same order, and Aggarwal [5] obtained the current record $U_c(n) \le n \log_2 n + 4n$ by exploiting the "diagonal property" and the "obtuse angle property" of distance matrices. For the multiplicity of the k-th largest distance, Altman [7, 8] proved $m_k \le kn$ in convex position; Moric and Pritchard [9] proved the Erdős–Lovász–Vesztergombi conjecture $m_1 + \cdots + m_k \le kn$ for $k \le 4$ and $m_1 + \cdots + m_k \le (2k-1)n$ in general, by computer search.

What this paper does. We do not solve Conjecture 1. We prove that it is *equivalent to a local degree statement* (Theorem 2), we prove a sharp structural description of the neighbourhood of every vertex (Theorem 1), we rule out one whole family of attacks by constructing arbitrarily many pairwise crossing unit distances (Theorem 3), and we prove bounded-density results for points on round/algrabraic sets. We also report a computational study, honestly recorded as inconclusive.

2 Definitions and three local facts

The following elementary observations drive everything.

Lemma 1 (common neighbours)**.** *If $x \neq y$ then $|N(x) \cap N(y)| \le 2$.*

Proof. $N(x) \cap N(y) \subseteq C(x,1) \cap C(y,1)$, and two distinct unit circles meet in at most two points. □

Lemma 2 (half-plane, angular monotonicity)**.** *Let P be strictly convex and $v \in P$.*

(a) *Every other point of P lies in the open half-plane bounded by the supporting line at v; hence $N(v)$ is contained in an arc of the circle $C(v,1)$ of angular length $< \pi$.*

(b) *The directions from v to the other points of P appear in the cyclic order of P.*

Proof. (a) is the definition of a supporting line at a vertex of a strictly convex set. For (b), the diagonals vw_i triangulate the polygon: the triangles $\triangle(vw_iw_{i+1})$ are pairwise interior-disjoint, which forces the directions $v \to w_1, \ldots, v \to w_{n-1}$ to be in cyclic order. □

Lemma 3 (the 60° lemma)**.** *If $x, y, z \in P$ are pairwise at distance 1, then $\triangle xyz$ is equilateral, so $\angle yxz = \pi/3$. In particular $G(P)$ contains no K_4.*

3 Neighbourhood structure

Theorem 1 (neighbourhoods)**.** *For every $v \in P$:*

(a) *$G[N(v)]$ is a disjoint union of paths whose components have at most three vertices;*

(b) *an edge of $G[N(v)]$ joins two unit-neighbours of v that are exactly $\pi/3$ apart as seen from v;*

(c) *v lies in at most $\frac{2}{3}\, d(v)$ unit triangles; hence P has at most $\frac{4}{9}\, u(P)$ unit triangles.*

Proof. By Lemma 2(a), $N(v)$ lies on an arc of $C(v,1)$ of length $< \pi$, and by Lemma 3 any edge of $G[N(v)]$ joins two points of that arc whose angular parameters differ by exactly $\pi/3$. So every vertex of $G[N(v)]$ has degree at most two (partners at $+\pi/3$ and $-\pi/3$, each present at most once). A component of maximum degree ≤ 2 is a path or a cycle; a component has a vertex of minimal angular parameter, which has degree at most one, so no component is a cycle, giving (b) and the first part of (a). A path with ℓ vertices has $\ell - 1$ edges, each spanning $\pi/3$ in angle; the whole

arc has length $< 3 \cdot \pi/3$, so at most three distinct angular values occur, giving $\ell \le 3$. Finally (c) follows from (a): the number of edges of $G[N(v)]$ is at most $\frac{2}{3}d(v)$, and each such edge is a unit triangle through v. Summing over v and dividing by 3 gives $\frac{4}{9}u(P)$. □

4 The bounded-degree reformulation

The following equivalence is the main conceptual contribution of the paper: it converts a global extremal question into a local one, and it tells us exactly *how much* progress any local improvement represents.

Theorem 2 (bridge)**.** *The following are equivalent.*

(1) $U_c(n) = O(n)$*;*

(2) *there is an absolute constant c such that every finite strictly convex point set has a vertex of unit-degree at most c;*

(3) *the class $\{G(P) : P$ finite, strictly convex$\}$ has bounded degeneracy.*

Proof. (2)⇒(1): by induction on n. Take a vertex v of degree $\le c$ and delete it; the remaining $n-1$ points are still strictly convex, so $u(P) \le c + u(P \setminus \{v\}) \le cn$.

(1)⇒(2): if $u(P) \le Cn$ then the average unit-degree is $2u(P)/n \le 2C$, so some vertex has degree at most $2C$; take $c = 2C$.

(2)⇔(3): every induced subgraph of $G(P)$ is the unit-distance graph of a subset of P, which is again strictly convex. □

Corollary 1. *Conjecture 1 is equivalent to the following local statement:* there is an absolute constant c such that every finite set of points in strictly convex position has a vertex incident to at most c unit segments. *The best statement presently available in the literature is the consequence of Aggarwal's bound $U_c(n) \le n \log_2 n + 4n$: some vertex has degree at most $2 \log_2 n + 8$.*

Note that Theorem 2(2) is strictly stronger than the Erdős–Fishburn bound $u(P) \le 2n$ only in the sense of its consequence: if $\delta(G(P)) \le c$ for an absolute c then $U_c(n) \le cn$, and with $c < 2$ this would already disprove nothing but would give a new upper bound; with $c = 1$ one would get $U_c(n) \le n$. Conversely $\delta(G(P)) \le 4$ for all P is *equivalent* to Erdős–Fishburn's $u(P) < 2n$, and $\delta(G(P)) \le 3$ implies $U_c(n) \le 3n$, which is a weaker but unconditional consequence of the 1959 conjecture. Thus the natural frontier of the local question is

$$\text{known: } \delta \le 2\log_2 n + 8, \qquad \text{target A: } \delta \le c \text{ (absolute)}, \qquad \text{target B: } \delta \le 3.$$

5 Pairwise crossing unit distances

A natural hope is that $G(P)$ is a topological graph with few crossings, and that a quasi-planarity theorem converts that into $O(n)$ edges. The next result closes that route.

Theorem 3 (crossing)**.** *For every $k \ge 1$ and every $n \ge 2k$ there is a set P of n points in strictly convex position for which $G(P)$ contains k pairwise crossing edges.*

Proof. Let $\delta = \pi/(3k)$ and put $2k$ points on the unit circle at angles $0, \delta, 2\delta, \dots, (2k-1)\delta$. They lie on a circular arc of angular length $(2k-1)\delta < 2\pi/3 < \pi$, hence are in strictly convex position. The k chords (z_i, z_{i+k}), $i = 1, \dots, k$, are the main diagonals of this convex $2k$-gon, and their endpoints occur in the cyclic order $z_i, z_j, z_{i+k}, z_{j+k}$ for $i < j$, so they cross in their interiors. Their length is

$$|z_{i+k} - z_i| = 2\sin\frac{k\delta}{2} = 2\sin\frac{\pi}{6} = 1.$$

For $n > 2k$, insert the extra points on the same circular arc at angles strictly between the existing ones; this preserves strict convexity and all unit distances already present. □

Corollary 2 (limitation)**.** *For no integer K can $U(n) = O(n)$ be deduced from a theorem of the form "a topological graph on n vertices with no $K+1$ pairwise crossing edges has at most $f(n)$ edges, with $f(n) = O(n)$".*

Proof. The bound $2n-4$ for topological graphs with no three pairwise crossing edges (Capoyleas–Pach) and its generalisations are vacuous here: taking $k > K$ in Theorem 3 produces a graph defeating the hypothesis. □

6 Round and algebraic sets

Proposition 1 (round sets)**.** *If the vertices of P lie on k circles then $u(P) \le kn$. More generally, if they lie on k algebraic curves of degree at most d, then $u(P) \le kdn$.*

Proof. A unit circle centred at a vertex v meets any circle in at most two points, and any curve of degree d in at most $2d$ points. Hence $d(v) \le 2kd$ and $u(P) = \frac{1}{2}\sum_v d(v) \le kdn$. □

In particular, points on a single circle give $u \le n$, and points on a parabola or on an ellipse give $u \le 2n$: the density predicted for convex position by the Erdős–Fishburn conjecture already holds for any such set. The input "two circles meet in at most two points" is classical (Erdős 1946), so this is best read as a remark about the class of sets for which the conjectured density is known to hold.

7 Computational work

All computations were run on 13 Oct 2026 with Python 3.14, `Fraction` for exact arithmetic and `numpy` for the numerical search. Two things were done.

(1) Exact certificates. Theorem 3 is certified exactly for $k = 2, 3, 4$ with rational coordinates: the $3/5 \times 4/5$ rectangle $(0,0), (\frac{3}{5}, 0), (\frac{3}{5}, \frac{4}{5}), (0, \frac{4}{5})$ has unit diagonals; the hexagon

$$(0,0), (\tfrac{3}{10}, -\tfrac{1}{2}), (\tfrac{7}{10}, -\tfrac{1}{2}), (1,0), (\tfrac{9}{10}, \tfrac{3}{10}), (\tfrac{1}{10}, \tfrac{3}{10})$$

and the analogous octagon have respectively three and four unit main diagonals. For each, `is_strictly_convex`, the unit property of every main diagonal, and the pairwise crossing of the main diagonals were verified in exact rational arithmetic (`tests/test_claims.py`, run: `python -m tests.test_claims`).

(2) An inconclusive search for density 2. We ran a simulated-annealing search (8000 iterations, 3 seeds, $n = 6, \dots, 14$) over convex n-gons, seeded both from jittered circles and from two-arc "lens" configurations (the family that realises $2n-7$), with a soft objective $\sum_{i<j} \exp(-(d_{ij} -$

$1)^2/2\sigma^2$) and annealed σ. The best densities found were $u/n \approx 0.67$ at $n = 6$ rising to ≈ 0.86 at $n = 14$ – an order of magnitude below the published $2 - 7/n$. No configuration with $u(P) \geq 2n$ was found. A separate grid-alignment search over two-circle lens configurations (solving the exact congruences that force all unit distances to lie on a common angular grid) produced at most $u = 9$ at $n = 9$, again below $2n - 7 = 11$.

Assessment. The negative outcome is honest but weak: the annealer does not even reach the known optimum, so it cannot be read as evidence for the Erdős–Fishburn bound. The computational study supports no claim and is included only as a reproducible record of an attempt.

8 What remains open

(a) Erdős–Moser 1959: $U_c(n) = O(n)$ (equivalently, by Theorem 2, a bounded local unit-degree).

(b) Erdős–Fishburn: $u(P) \leq 2n$ (equivalently $\delta(G(P)) \leq 4$).

(c) Erdős #97: no finite convex point set has property E_4 (every point has ≥ 4 others at a common distance).

(d) Whether $\delta(G(P)) \leq 3$ for all finite convex P, which would give $U_c(n) \leq 3n$ and settle (a) with a constant.

(e) Improving $\log_2 n + 4$ to $o(\log n)$ in Aggarwal's bound; our structural lemmas did not suffice. In particular, the forbidden 2×2 all-ones submatrix of cut matrices that a diagonal split cannot contain is the pattern S_2 of Fishburn–Reeds [6], so that observation carries no novelty.

We also recorded, but could not exploit, the following: Lemma 1 together with Theorem 1 permits minimum degrees as large as $\Theta(\sqrt{n})$ (the bound $\sum_v \binom{d(v)}{2} = \sum_{x \neq y} |N(x) \cap N(y)| \leq 2\binom{n}{2}$ is satisfied by $d(v) \approx 2\sqrt{n}$), so a constant degeneracy bound must come from the geometry, not from counting alone.

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This is an autonomous research run; no external human input shaped the mathematics. All literature checks were made against the live arXiv API and publisher pages on 13 Oct 2026; Aggarwal's paper was read in full before the novelty assessment was finalised.

References

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